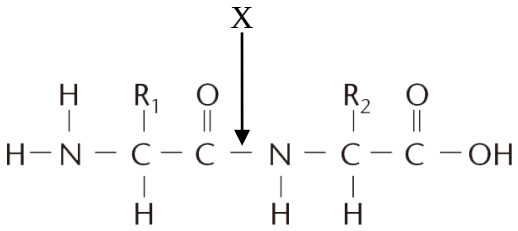
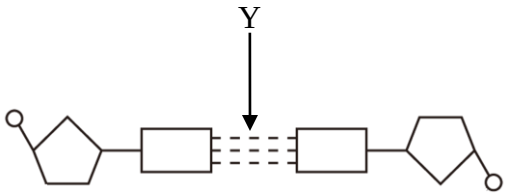
Multiple-choice questions

|!|BCEMC0020000001|!|

The diagram below shows two types of molecules.



dipeptide



two nucleotides

Which of the following combinations correctly identifies bond X and bond Y?

	bond X	***bond Y***
A.	hydrogen bond	hydrogen bond
B.	peptide bond	hydrogen bond
C.	peptide bond	peptide bond
D.	hydrogen bond	peptide bond

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000002|!|

Molecules X and Y are amino acids, and Z is a dipeptide. Which of the following reactions represent a condensation reaction?

A. $Z + H_2O \rightarrow X + Y$

B. $Z \rightarrow X + Y + H_2O$

C. $X + Y \rightarrow Z + H_2O$

D. $X + Y + H_2O \rightarrow Z$

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0020000003|!|

Which of the following molecules contains carbon, hydrogen and oxygen only?

(1) carbohydrate
(2) protein
(3) lipid

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000004|!|

Which characteristic(s) of water allow(s) effective transport of nutrients around the body by blood?

(1) Water is a good solvent.
(2) Water takes away heat when it evaporates.
(3) Water is transparent.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0020000005|!| ★

Which of the following is a function of sucrose in plants?

A. for transport
B. for support
C. for protection
D. for photosynthesis

##A##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMB0020000006|!| ★

Which of the following are all protein in nature?

A. hormones, enzymes and triglycerides
B. nitrates, antibodies and receptors
C. enzymes, antibodies and receptors
D. nitrates, triglycerides and enzymes

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0020000007|!| ★

Why is glycogen a better storage molecule than glucose?

A. Glycogen can release energy.
B. Glycogen can release energy faster than its individual glucose molecules.
C. Glycogen takes up less space than its individual glucose molecules.
D. Glycogen is only stored in the liver.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0020000008|!| ★

A person with very low blood sugar level is advised to drink a glass of glucose solution instead of eating a piece of bread. Which of the following is a correct explanation for this?

A. Bread contains starch which is insoluble in water.
B. Starch in bread needs some time to be digested.
C. Glucose solution contains more water than the bread.
D. Glucose solution contains more energy than the bread.

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0020000009|!| ★

For the following pairs of carbohydrates, we can use the Benedict's test to distinguish the two carbohydrates from one another except

(1) glucose and galactose
(2) glucose and maltose
(3) glucose and starch
(4) glucose and sucrose

A. (1) and (2)
B. (1) and (3)
C. (2) and (4)
D. (3) and (4)

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0020000010|!|

Which of the following substances make up the backbone of a polynucleotide of an RNA molecule?

(1) ribose
(2) deoxyribose
(3) phosphate
(4) phospholipid
(5) nitrogenous base

A. (1) and (3) only
B. (2) and (3) only
C. (1), (4) and (5) only
D. (2), (3) and (5) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0020000011|!| ★

Which of the following is a carbohydrate?

A. $CaCO_3$

B. C_4H_9OH

C. $CH_3(CH_2)_{12}COOH$

D. $C_8H_{16}O_8$

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0020000012|!|

Which of the following is ***not*** made up of glucose units?

A. lipids

B. glycogen

C. cellulose

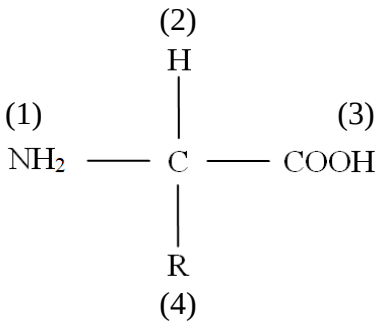
D. starch

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000013|!| ★

An amino acid molecule has the following structure:



Which two groups combine to form a peptide link?

A. (1) and (2)

B. (1) and (3)

C. (2) and (4)

D. (2) and (3)

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000014|!|

Which of the following statements about the properties of water is ***incorrect***?

A. Evaporation of water is effective to help cool the body of an organism.
B. Large volumes of water are slow to change temperature as the environmental temperature changes.
C. Ice (solid form of water) is denser than liquid water.
D. Water is a good solvent for ions and simple sugars

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0020000015|!| ★★

The basic units of nucleic acids are nucleotides. Each nucleotide is made up of a 5-carbon sugar, a base and a phosphate. In a DNA molecule, what is the maximum number of components to which a 5-carbon sugar is linked?

A. one phosphate and one base
B. two phosphates and one base
C. one phosphate and two bases
D. two phosphates and two bases

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0020000016|!|

Which of the following reactions occurs when a dipeptide is formed from two amino acids?

A. oxidation
B. reduction
C. hydrolysis
D. condensation

##D##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000017|!|

Which of the following molecules contains the element phosphorus?

A. amino acids

B. fatty acids

C. ribose

D. nucleotides

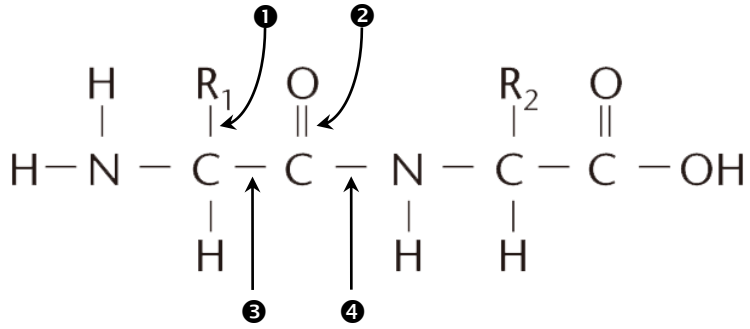
□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000018|!|

Which of the following represents the peptide linkage of a dipeptide?



A. ❶

B. ❷

C. ❸

D. ❹

□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0020000019|!| ★★

A student performed a Benedict's test on two brands of coke, A and B. Coke A contains more sugars than coke B. 1 cm^3 of Benedict's solution and 3 cm^3 of coke A were mixed in a test tube and then heated in a water bath. The same steps were repeated using coke B. The amount of precipitate measured for the two brands of coke was the same. What is a possible source of error that causes the results?

A. The mixture should not be heated in a water bath.
B. The test tubes containing Benedict's solution and coke were not put into a water bath before mixing.
C. The amount of Benedict's solution was not enough.
D. The two samples of coke contained the same amount of sugars.

##C##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0020000020|!| ★

The table below shows the number of carbon, hydrogen and oxygen atoms in four biomolecules:

	Number of atoms		
Biomolecule	**carbon**	**hydrogen**	**oxygen**
W	6	12	6
X	600	1000	500
Y	27	46	0
Z	6	8	6

Which biomolecule(s) may show a positive result in Benedict's test?

A. W only
B. Y only
C. W and X only
D. Y and Z only

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0020000021|!| ★

A student carried out Benedict's test and iodine test on a food sample. The table below shows the results.

Benedict's test	Iodine test
Formation of brick-red precipitates	Iodine solution turns blue-black

This sample must contain

(1) glucose.
(2) sucrose.
(3) starch.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0020000022|!| ★

DNA consists of four bases. If the percentage of guanine in the DNA of an organism is approximately 30%, what is the percentage of adenine?

A. 20 %
B. 30 %
C. 40 %
D. 70 %

##A##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMC0020000023|!|

Which of the following is / are the functions of proteins in the human body?

(1) making antibodies
(2) making the cell membrane
(3) providing energy

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000024|!|

Which of the following statements about DNA and RNA is ***incorrect***?

A. DNA is a double helix while RNA is a single chain.
B. Uracil can only be found in RNA but not in DNA.
C. Both DNA and RNA contain ribose sugar.
D. The nucleotides in the polynucleotide chains of both DNA and RNA are joined by condensation.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000025|!| ★

Which of the following substances are organic?

(1) glucose
(2) calcium phosphate
(3) haemoglobin

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000026|!| ★

Which of the following carbon-containing substances is inorganic?

A. adenine
B. triose phosphate
C. carbon dioxide
D. triglyceride

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000027|!| ★

Which of the following combinations about the properties of water and its corresponding function in organisms are correct?

	Property of water	***Function of water***
(1)	good solvent	acts as a reactant in photosynthesis
(2)	high latent heat of vaporization	acts as a cooling agent
(3)	high density	provides buoyancy for aquatic plants and animals

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0020000028|!| ★

Which of the following is ***not*** a function of water in organisms?

A. supports non-woody plants when plant cells take in sufficient water and become turgid
B. provides a medium for the transport of dissolved nutrients and metabolic wastes in the human body
C. lowers the human body temperature when water in sweat evaporates
D. is used in condensation reactions to synthesize macromolecules like polypeptides

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0020000029|!|

The following equation shows the process of photosynthesis in green plants.

$$\text{water} + \text{carbon dioxide} \xrightarrow[\text{chlorophyll}]{\text{light}} \text{carbohydrates} + \text{oxygen}$$

Which role(s) does water play in the above process?

(1) as a reactant
(2) as a reaction medium
(3) as a cooling agent

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0020000030|!| **Cross topic**

What is the function of the water film on the inner surface of the air sacs in the lungs?

A. to act as a reactant for respiration
B. to allow respiratory gases to be dissolved
C. to protect the surface against infections
D. to provide a cooling effect

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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000031|!|

Which of the following is the function of iron in mammals?

A. forming haemoglobin
B. forming cell membrane
C. for muscle contraction
D. forming bones

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000032|!| ★

Which of the following parts of a plant cell contain phosphate?

(1) DNA in the nucleus
(2) cell membrane
(3) cell wall

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000033|!|

Which of the following combinations about the function of the mineral is ***incorrect***?

	Mineral	***Function***
A.	calcium	for forming teeth
B.	magnesium	for blood clotting
C.	phosphate	for forming nucleic acids
D.	iron	for forming haemoglobin

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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000034|!|

A disaccharide molecule can be broken down into two molecules of monosaccharide through

A. condensation.
B. hydrolysis.
C. photolysis.
D. deamination.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000035|!|

Which of the following carbohydrates is insoluble in water?

A. fructose
B. sucrose
C. maltose
D. cellulose

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0020000036|!| ★★

Which of the following substances give(s) a positive result in the Benedict's test?

(1) glucose
(2) deoxyribose
(3) starch

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000037|!|

In a galactose molecule, the hydrogen and oxygen atoms are in the ratio of

A. 1 : 1.
B. 1 : 2.
C. 2 : 1.
D. 6 : 1.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000038|!|

Which of the following sugars does ***not*** give a positive result in the Benedict's test?

A. glucose
B. fructose
C. galactose
D. sucrose

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000039|!|

Which of the following is/are the function(s) of lipids in organisms?

(1) as a component of cell membranes
(2) as a form of energy stores in plants
(3) as an heat insulator in mammals

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000040|!|

Which of the following is ***not*** a component of a phospholipid molecule?

A. fatty acid
B. glycerol
C. phosphate
D. cholesterol

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000041|!|

How many fatty acid molecules are present in a phospholipid molecule?

A. 1
B. 2
C. 3
D. 4

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000042|!| ★

Energy can be stored as glycogen and adipose tissues in our bodies. Which of the following correctly shows their locations in our bodies?

	Glycogen	***Adipose tissues***
A.	under the skin	around internal organs
B.	in muscles	under the skin
C.	in the liver	in muscles
D.	around internal organs	in the liver

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0020000043|!| ★

Which of the following substances contain(s) nitrogen?

(1) antibody
(2) steroid
(3) haemoglobin

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000044|!| ★

Which of the following combinations about examples of proteins and lipids is correct?

	Proteins	***Lipids***
A.	haemoglobin	insulin
B.	enzyme	glycogen
C.	DNA	cholesterol
D.	antibody	triglyceride

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000045|!| ★

Which of the following statements about proteins are correct?

(1) Amino acids are linked by hydrogen bonds in a protein molecule.
(2) Proteins are present in the cell membrane of both prokaryotic and eukaryotic cells.
(3) Enzymes are denatured when they are exposed to a high temperature because of their protein nature.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000046|!| ★

How many water molecules are required to completely hydrolyse a polypeptide composed of 15 amino acids?

A. 14
B. 15
C. 16
D. 30

##A##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMC0020000047|!|

Which of the following substances contain(s) nitrogen?

(1) RNA
(2) glucose
(3) polypeptide

A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000048|!|

Which of the following statements about DNA molecules is ***incorrect***?

A. They are found mainly in the cytoplasm.
B. They are replicated and passed to the offspring in asexual reproduction.
C. They determine the types of proteins produced.
D. They contain thymine but no uracil.

##A##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000049|!|

Which of the following are the correct base pairs in a DNA molecule?

A. G—T, A—C

B. G—C, A—T

C. T—U, A—G

D. G—C, A—U

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000050|!|

Which of the following nitrogenous bases is ***not*** included in a DNA molecule?

A. thymine

B. uracil

C. cytosine

D. guanine

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000051|!| ★

Which of the following statements about DNA is ***incorrect***?

A. DNA does not contain oxygen.

B. DNA contains thymine.

C. DNA is found in the nucleus of a eukaryotic cell.

D. The sugar in a DNA molecule is a five-carbon sugar.

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0020000052|!| ★

In the DNA of a grasshopper, 20% of the nitrogenous bases are cytosine (C). What is the ratio of adenine (A) to thymine (T) in the DNA of a grasshopper?

A. 1 : 1
B. 2 : 1
C. 2 : 3
D. 3 : 2

##A##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0020000053|!| ★

Which of the following comparisons between RNA and DNA is ***incorrect***?

	RNA	***DNA***
A.	contains ribose	contains deoxyribose
B.	single-stranded form	double helix form
C.	does not contain the base adenine	does not contain the base uracil
D.	contains the base uracil	contains the base thymine

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000054|!|

Which of the following is found in both DNA and RNA?

A. ribose
B. uracil
C. phosphate group
D. amino group

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0020000055|!| ★

Which of the following organic substances can be used as energy storage(s) in plants?

(1) glycogen
(2) triglyceride
(3) starch

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000056|!|

Which of the following substances can be used to provide energy if carbohydrates and lipids are used up by the body?

(1) proteins
(2) nucleic acids
(3) minerals

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000058|!|

Carbohydrates are usually stored in plants as

A. glycogen.
B. cellulose.
C. starch.
D. glucose.

##C##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0020000059|!| ★

Which of the following are the most abundant organic substances in the human body?

A. carbohydrates
B. nucleic acids
C. proteins
D. minerals

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000060|!|

Which of the following substances is directly used by human cells in respiration to release energy?

A. glucose
B. glycogen
C. amino acid
D. glycerol

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0020000061|!|

Which of the following is the basic unit of DNA?

A. deoxyribose
B. nucleotide
C. amino acid
D. nitrogenous base

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0020000062|!| ★

Which of the following show that water is a supporting agent?

(1) rigidity of xylem vessels in trees
(2) buoyancy of aquatic plants
(3) the hydrostatic skeleton in earthworms

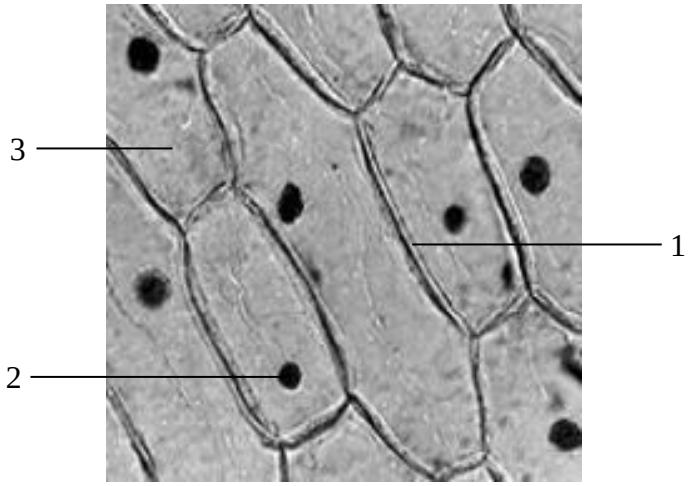
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0020000064|!| ★

Directions: The following two questions refer to the photomicrograph below, which shows some onion epidermal cells.



RNA can be found in

A. 2 only.
B. 1 and 2 only.
C. 1 and 3 only.
D. 2 and 3 only.

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0020000065|!| ★

Which of the following combinations about the major components of structures 1 and 3 is correct?

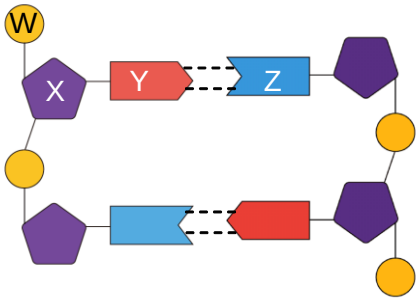
	Structure 1	***Structure 3***
A.	cellulose	water
B.	cellulose	protein
C.	lipid	water
D.	lipid	protein

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0020000066|!| ★

Directions: The following two questions refer to the diagram below which shows two base pairs of a DNA molecule.



Which of the following combinations correctly identifies subunits W, X and Y?

	W	***X***	***Y***
A.	nitrogenous base	deoxyribose	phosphate group
B.	phosphate group	deoxyribose	nitrogenous base
C.	deoxyribose	phosphate group	nitrogenous base
D.	phosphate group	nitrogenous base	deoxyribose

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0020000067|!| ★

Which of the following correctly identifies the base pair represented by subunits Y and Z?

A. A – T
B. U – A
C. G – T
D. A – C

□

##A##

Skill assessed	Mark
Analysing data	1

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